Optimal Control and Reinforcement Learning
Nonlinear Optimization

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The present slides are for internal use of the course
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Unconstrained optimization: definitions, optimality conditions and special problem classes

Unconstrained optimization: introduction

Consider the unconstrained optimization problem

$$\min_{z \in \mathbb{R}^d} \ell(z),$$

with $\ell : \mathbb{R}^d \to \mathbb{R}$ a cost function to be minimized and z a decision vector.

We say that $z^\star$ is a

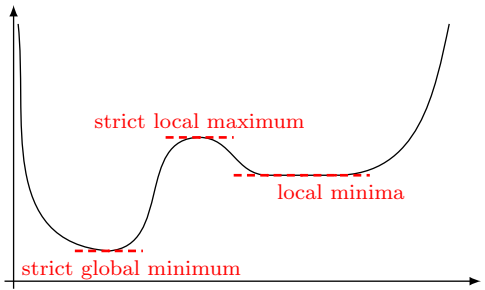
- global minimum if $\ell(z^\star) \leq \ell(z)$ for all $z \in \mathbb{R}^d$
- strict global minimum if $\ell(z^\star) < \ell(z)$ for all $z \neq z^\star$
- local minimum if there exists $\epsilon > 0$ such that $\ell(z^\star) \leq \ell(z)$ for all $z \in B(z^\star, \epsilon) = \{z \in \mathbb{R}^d \mid \|z - z^\star\| < \epsilon\}$
- strict local minimum if there exists $\epsilon > 0$ such that $\ell(z^\star) < \ell(z)$ for all $z \in B(z^\star, \epsilon)$ and $z \neq z^\star$.

Unconstrained optimization: introduction

Consider the unconstrained optimization problem

$$\min_{z \in \mathbb{R}^d} \ell(z),$$

with $\ell : \mathbb{R}^d \to \mathbb{R}$ a cost function to be minimized and z a decision vector.



Remark Maxima can be equivalently defined. Moreover, maxima are minima of $-\ell$.

Notation

Optimal solution

We denote $\ell(z^*)$ the optimal (minimum) value of a generic optimization problem, i.e.

$$\ell(z^*) = \min_{z \in \mathbb{R}^d} \ell(z),$$

where z^* is the minimum point (optimal value for the optimization variable), i.e.,

$$z^* = \operatorname*{argmin}_{z \in \mathbb{R}^d} \ell(z),$$

Remarks on notation: gradient and Hessian matrix of a function

Gradient of a function

For a function $r : \mathbb{R}^d \to \mathbb{R}$, the gradient is denoted as

$$\nabla r(z) = \begin{bmatrix} \frac{\partial r(z)}{\partial z_1} \\ \vdots \\ \frac{\partial r(z)}{\partial z_d} \end{bmatrix} \in \mathbb{R}^{d \times 1}$$

Hessian matrix of a function

For a function $r : \mathbb{R}^d \to \mathbb{R}$, the Hessian matrix is denoted as

$$\nabla^2(r(z)) = \begin{bmatrix} \frac{\partial^2 r(z)}{\partial z_1^2} & \cdots & \frac{\partial^2 r(z)}{\partial z_1 z_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial r(z)}{\partial z_d z_1} & \cdots & \frac{\partial^2 r(z)}{\partial z_d^2} \end{bmatrix} \in \mathbb{R}^{d \times d}$$

The Hessian matrix is a symmetric matrix, since the assumption of continuity of the second derivatives implies that the order of differentiation does not matter.

Remarks on notation: gradient of a vector-valued function

Gradient of a (vector-valued) function

For a vector field $r : \mathbb{R}^d \to \mathbb{R}^m$, the gradient is denoted as

$$\begin{aligned}\nabla r(z) &= \begin{bmatrix} \nabla r_1(z) & \cdots & \nabla r_m(z) \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial r_1(z)}{\partial z_1} & \cdots & \frac{\partial r_m(z)}{\partial z_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial r_1(z)}{\partial z_d} & \cdots & \frac{\partial r_m(z)}{\partial z_d} \end{bmatrix} \in \mathbb{R}^{d \times m}\end{aligned}$$

which is the transpose of the Jacobian matrix of r.

Necessary conditions of optimality (unconstrainted)

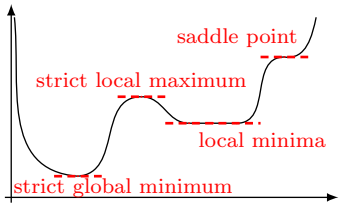
First order necessary condition (FNC) of optimality (unconstrained)

Let $z^\star$ be an unconstrained local minimum of $\ell : \mathbb{R}^d \to \mathbb{R}$ and assume that ℓ is continuously differentiable ($\mathcal{C}^1$) in $B(z^\star, \epsilon)$ for some $\epsilon > 0$. Then $\nabla \ell(z^\star) = 0$.

Second order necessary condition of optimality (unconstrained)

If additionally ℓ is twice continuously differentiable ($\mathcal{C}^2$) in $B(z^\star, \epsilon)$, then $\nabla^2 \ell(z^\star) \geq 0$ (the Hessian $\nabla^2 \ell(z^\star)$ of ℓ is positive semidefinite).

Remark Points $\bar{z}$ satisfying $\nabla \ell(\bar{z}) = 0$ are called stationary points. They include minima, maxima and saddle points.



Second order sufficient conditions of optimality

Second order sufficient condition of optimality (unconstrained)

Let $\ell : \mathbb{R}^d \to \mathbb{R}$ be twice continuously differentiable ($\mathcal{C}^2$) in $B(z^\star, \epsilon)$ for some $\epsilon > 0$. Suppose that $z^\star \in \mathbb{R}^d$ satisfies

$$\nabla \ell(z^\star) = 0 \quad \text{and} \quad \nabla^2 \ell(z^\star) > 0 \text{ (positive definite).}$$

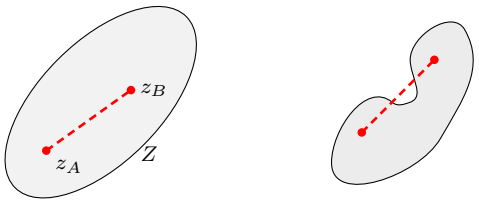
Then $z^\star$ is a strict (unconstrained) local minimum of ℓ.

Convex sets

Convex set

A set $Z \subset \mathbb{R}^d$ is convex if for any two points z_A and z_B in Z and for all $\theta \in [0, 1]$, then

$$\theta z_A + (1 - \theta) z_B \in Z.$$



Convex set

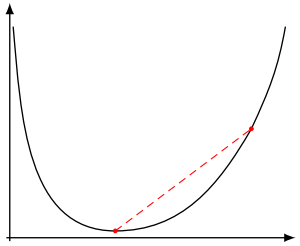
Nonconvex set

Convex functions

Convex function

Let $Z \subset \mathbb{R}^d$ be a convex set. A function $\ell : Z \to \mathbb{R}$ is convex if for any two points z_A and z_B in Z and for all $\theta \in [0, 1]$, then

$$\ell(\theta z_A + (1 - \theta) z_B) \leq \theta \ell(z_A) + (1 - \theta) \ell(z_B).$$



Remark A function ℓ is concave if $-\ell$ is convex. A function ℓ is strictly convex if the inequality holds strictly for $z_A \neq z_B$ and $\theta \in (0, 1)$.

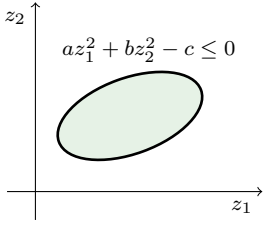
Inequality constraints and convex sets

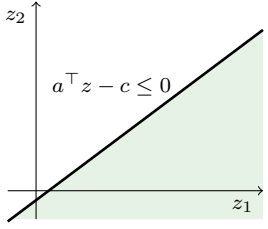
Let $g : \mathbb{R}^d \to \mathbb{R}^p$, we can define a set $Z_{\text{ineq}} \subset \mathbb{R}^d$ as

$$Z_{\text{ineq}} = \{z \in \mathbb{R}^d \mid g(z) \leq 0\}$$

The set Z_{ineq} is convex if and only if g is a quasi-convex function (e.g., monotone functions on the axis).

Corollary: If g is convex, then Z_{ineq} is convex.



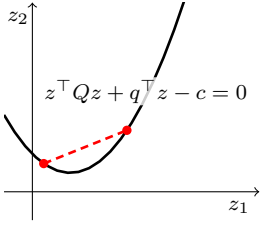


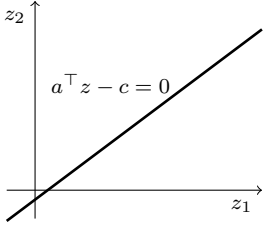
Equality constraints and convex sets

Let $h : \mathbb{R}^d \to \mathbb{R}^p$, we can define a set $Z_{\text{eq}} \subset \mathbb{R}^d$ as

$$Z_{\text{eq}} = \{z \in \mathbb{R}^d \mid h(z) = 0\}$$

The set Z_{eq} is convex if and only if h is an affine function. Convex sets identified through equality constraints are linear spaces (hyperplanes).





Convexity and gradient monotonicity

If ℓ is differentiable and convex, then for all $z_A, z_B \in Z$ it holds

$$\ell(z_B) \geq \ell(z_A) + \nabla \ell(z_A)^\top (z_B - z_A)$$

If ℓ is differentiable and convex, then its gradient $\nabla \ell : \mathbb{R}^d \to \mathbb{R}^d$ satisfies

$$\Big(\nabla \ell(z_A) - \nabla \ell(z_B)\Big)^\top (z_A - z_B) \geq 0$$

for all $z_A, z_B \in Z$.

That is, the gradient $\nabla \ell$ is a monotone operator

Strict convexity and gradient monotonicity

A function ℓ is strictly convex if for $z_A \neq z_B$ and $\theta \in (0,1)$

$$\ell(\theta z_A + (1-\theta) z_B) < \theta \ell(z_A) + (1-\theta)\ell(z_B)$$

If the strictly convex function ℓ is also differentiable, then its gradient satisfies

$$\Big(\nabla \ell(z_A) - \nabla \ell(z_B)\Big)^\top (z_A - z_B) > 0$$

for all $z_A, z_B \in Z$.

That is, the gradient $\nabla \ell$ is a strictly monotone operator

Strong convexity and gradient monotonicity

A function ℓ is strongly convex with parameter $\mu > 0$ if for $z_A \neq z_B$ and $\theta \in (0, 1)$

$$\ell(\theta z_A + (1-\theta) z_B) < \theta \ell(z_A) + (1-\theta)\ell(z_B) - \mu\theta(1-\theta)\|z_A - z_B\|^2$$

The gradient of a differentiable strongly convex function satisfies

$$\Big(\nabla \ell(z_A) - \nabla \ell(z_B)\Big)^\top (z_A - z_B) \geq \mu \|z_A - z_B\|^2$$

for all $z_A, z_B \in Z$.

That is, the gradient $\nabla \ell$ is a strongly monotone operator

Convexity and Lipschitz continuity of the gradient

Consider a differentiable convex function ℓ with a Lipschitz continuous gradient with parameter $L > 0$, i.e.,

$$\|\nabla \ell(z_A) - \nabla \ell(z_B)\| \leq L \|z_A - z_B\|$$

for all $z_A, z_B \in Z$.

Then, the following characterization holds

$$\Big(\nabla \ell(z_A) - \nabla \ell(z_B)\Big)^\top (z_A - z_B) \geq \tfrac{1}{L} \|\nabla \ell(z_A) - \nabla \ell(z_B)\|^2$$

for all $z_A, z_B \in Z$.

That is, the gradient $\nabla \ell$ is a co-coercive operator

Strong convexity and Lipschitz continuity of the gradient

Consider a strongly convex (with parameter $\mu > 0$) function ℓ with Lipschitz continuous gradient (with parameter $L > 0$)

Then the following characterization holds

$$\Big(\nabla \ell(z_A) - \nabla \ell(z_B)\Big)^\top (z_A - z_B) \geq \underbrace{\tfrac{\mu L}{\mu + L}}_{\gamma_1} \|z_A - z_B\|^2 + \underbrace{\tfrac{1}{\mu + L}}_{\gamma_2} \|\nabla \ell(z_A) - \nabla \ell(z_B)\|^2$$

for all $z_A, z_B \in Z$.

Graphical interpretation of convexity

If ℓ is differentiable and convex, then

$$\ell(z_B) \geq \ell(z_A) + \nabla \ell(z_A)^\top (z_B - z_A)$$

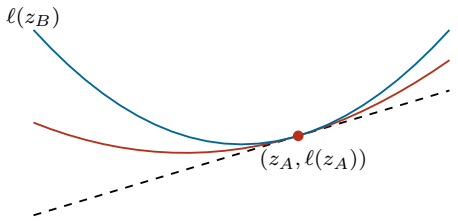
for all $z_A, z_B \in Z$.

If ℓ is differentiable and μ-strongly convex, then

$$\ell(z_B) \geq \ell(z_A) + \nabla \ell(z_A)^\top (z_B - z_A) + \frac{\mu}{2} \|z_B - z_A\|^2$$

for all $z_A, z_B \in Z$.

That is, strong convexity imposes a quadratic lower bound on ℓ



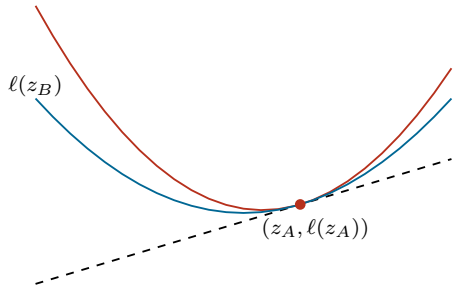
Graphical interpretation of Lipschitz continuity of the gradient

If ℓ is differentiable with Lipschitz continuous gradient, then

$$\ell(z_B) \leq \ell(z_A) + \nabla\ell(z_A)^\top (z_B - z_A) + \tfrac{L}{2}\|z_B - z_A\|^2$$

for all $z_A, z_B \in Z$.

That is, Lipschitz continuity of $\nabla\ell$ imposes a quadratic upper bound on ℓ



Minimization of convex functions

Proposition (Prop. B.10 Nonlin. Progr.)
Let $Z \subset \mathbb{R}^d$ be a convex set and $\ell : Z \to \mathbb{R}$ a convex function. Then a local minimum of ℓ is also a global minimum.

Proof
Let $z^\star$ be a local minimum of ℓ. Suppose by contradiction that $z^\star$ is not a global minimum. Then there exists a global minimum $\bar{z} \in X$ such that $\ell(\bar{z}) < \ell(z^\star)$.

Using the definition of convexity it holds

$$\begin{aligned}\ell(\theta z^\star + (1-\theta)\bar{z}) &\leq \theta\ell(z^\star) + (1-\theta)\ell(\bar{z}) \\ &< \theta\ell(z^\star) + (1-\theta)\ell(z^\star) = \ell(z^\star),\end{aligned}$$

for every $\theta \in [0, 1)$.

This contradicts the assumption that $z^\star$ is a local minimum since we found points, $\tilde{z} = \theta z^\star + (1-\theta)\bar{z}$ with $\theta \in [0, 1)$, arbitrarily close to $z^\star$ with $\ell(\tilde{z}) < \ell(z^\star)$. This concludes the proof.

Minimization of convex functions

Necessary and sufficient condition of optimality (unconstrained)

For the unconstrained minimization of a convex function it can be shown that the first order necessary condition of optimality is also sufficient (for a global minimum).

Proposition (Prop. 1.1.2 Nonlin. Progr.)

Let $\ell : \mathbb{R}^d \to \mathbb{R}$ be a convex function. Then $z^\star$ is a global minimum if and only if $\nabla \ell(z^\star) = 0$.

Proof (sketch)

Use the fact (Proposition B3 in Nonlinear Programming) that ℓ convex if and only if

$$\ell(z) \geq \ell(z^\star) + \nabla \ell(z^\star)^T (z - z^\star)$$

for all $z \in \mathbb{R}^d$. Thus, if $\nabla \ell(z^\star) = 0$ it follows immediately $\ell(z) \geq \ell(z^\star)$ for all $z \in \mathbb{R}^d$. The converse is true by the first order necessary condition and the fact that a local minimum is a global minimum.

Quadratic Programming (unconstrained)

Let us consider a special class of optimization problems, namely quadratic optimization problems or quadratic programs.

Quadratic program

$$\min_{z \in \mathbb{R}^d} \ \frac{1}{2} z^T Q z + q^T z$$

with $Q = Q^T \in \mathbb{R}^{d \times d}$ and $q \in \mathbb{R}^d$.

Quadratic Programming (unconstrained): optimality conditions

First-order necessary condition for optimality: if $z^\star$ is a minimum then

$$\nabla \ell(z^\star) = 0 \implies Qz^\star + q = 0$$

Second-order necessary condition for optimality: if $z^\star$ is a minimum then

$$\nabla^2 \ell(z^\star) \geq 0 \implies Q \geq 0$$

IMPORTANT

A necessary condition for the existence of minima for a quadratic program is that $Q \geq 0$.
Thus, quadratic programs admitting at least a minimum are convex optimization problems.

Quadratic Programming (unconstrained): properties

Since quadratic programs are convex program ($Q \geq 0$ is necessary to have a local minimum), then the following holds.

IMPORTANT

For a quadratic program necessary conditions of optimality are also sufficient and minima are global.

If $Q > 0$, then there exists a unique global minimum given by

$$z^\star = -Q^{-1}q.$$

Unconstrained optimization: algorithms

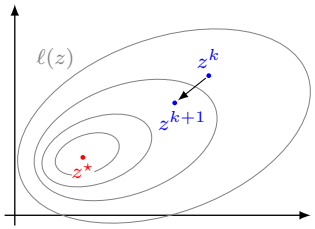
Optimization algorithms (unconstrained): iterative descent methods

We consider optimization algorithms relying on the iterative descent idea.

Notation We denote $z^k \in \mathbb{R}^d$ an estimate of a local minimum at iteration $k \in \mathbb{N}$.

The algorithm starts at a given initial guess z^0 and iteratively generates vectors $z^1, z^2, \ldots$ such that ℓ is decreased at each iteration, i.e.,

$$\ell(z^{k+1}) < \ell(z^k) \qquad k = 1, 2, \ldots$$



Optimization algorithms: two-step procedure

We consider a general two-step procedure that reads as follows

$$z^{k+1} = z^k + \gamma^k d^k, \qquad k = 1, 2, \dots$$

in which

1. each $\gamma^k > 0$ is a "step-size",
2. $d^k \in \mathbb{R}^d$ is a "direction".

The goal is to

1. choose a direction d^k along which the cost decreases for γ^k sufficiently small;
2. select a step-size γ^k guaranteeing a sufficient decrease.

Notation In other references these are called line-search methods.

Gradient methods: introduction

Let z^k be such that $\nabla \ell(z^k) \neq 0$. We start by considering the update rule

$$z^{k+1} = z^k - \gamma^k \nabla \ell(z^k),$$

i.e., we choose $d^k = -\nabla \ell(z^k)$.

From the first order Taylor expansion of ℓ at z^k we have

$$\begin{aligned} \ell(z^{k+1}) &= \ell(z^k) + \nabla \ell(z^k)^T (z^{k+1} - z^k) + o(\|z^{k+1} - z^k\|) \\ &= \ell(z^k) - \gamma^k \|\nabla \ell(z^k)\|^2 + o(\gamma^k). \end{aligned}$$

Thus, for $\gamma^k > 0$ sufficiently small it can be shown that $\ell(z^{k+1}) < \ell(z^k)$.

Gradient methods: introduction

The update rule

$$z^{k+1} = z^k - \gamma^k \nabla \ell(z^k),$$

can be generalized to so called gradient methods

$$z^{k+1} = z^k + \gamma^k d^k,$$

with d^k such that

$$\nabla \ell(z^k)^T d^k < 0.$$

Gradient methods: selecting the descent direction

Several gradient methods can be written as

$$z^{k+1} = z^k - \gamma^k D^k \nabla \ell(z^k) \qquad k = 1, 2, \dots$$

where $D^k \in \mathbb{R}^{d \times d}$ is a symmetric positive definite matrix.
It can be immediately seen that

$$-\nabla \ell(z^k)^T D^k \nabla \ell(z^k) < 0,$$

i.e., $d^k = -D^k \nabla \ell(z^k)$ is a descent direction.

Gradient methods: selecting the descent direction

Some choices of D^k

- Steepest descent $D^k = I_d$
- Newton's method $D^k = \left(\nabla^2 \ell(z^k)\right)^{-1}$

  It can be used when $\nabla^2 \ell(z^k) > 0$. It typically converges very fast asymptotically. For $\gamma^k = 1$ pure Newton's method.
- Discretized Newton's method $D^k = \left(H(z^k)\right)^{-1}$,

  where $H(z^k)$ is a positive definite symmetric approximation of $\nabla^2 \ell(z^k)$ obtained by using finite difference approximations of the second derivatives.

Steepest descent (gradient method)

The update rule obtained for $D^k = I$,

$$z^{k+1} = z^k - \gamma^k \nabla \ell(z^k),$$

is called steepest descent (or gradient method tout court).

The name steepest descent is due to the following property. The normalized negative gradient direction

$$d^k = -\frac{\nabla \ell(z^k)}{\|\nabla \ell(z^k)\|},$$

minimizes the slope $\nabla \ell(z^k)^T d^k$ among all normalized directions, i.e., it gives the steepest descent.

Newton's method for root finding

Consider the nonlinear root (zero) finding problem

$$r(z) = 0$$

Idea: Iteratively refine the solution such that the improved guess z^{k+1} represents a root (zero) of the linear approximation of r about the current tentative solution z^k.

Consider the linear approximation of r about z^k, we have

$$r^k(z^k + \Delta z^k) = r(z^k) + \nabla r(z^k)^\top \Delta z^k$$

then, finding the zeros of the approximation, we have

$$\Delta z^k = -\left(\nabla r(z^k)^\top\right)^{-1} r(z^k)$$

Thus, the solution is improved as

$$z^{k+1} = z^k - \left(\nabla r(z^k)^\top\right)^{-1} r(z^k).$$

Newton's Method for Unconstrained Optimization

Consider the unconstrained optimization problem

$$\min_{z \in \mathbb{R}^d} \ell(z),$$

stationary points $\bar{z}$ satisfy the first-order optimality condition

$$\nabla \ell(\bar{z}) = 0.$$

We can look at it as a root finding problem, with $r(z) = \nabla \ell(z)$, and solve it via Newton's method.

Therefore, we can compute Δz^k as the solution of the linearization of $r(z) = \nabla \ell(z)$ at z^k, i.e.,

$$\nabla \ell(z^k) + \nabla^2 \ell(z^k) \Delta z^k = 0,$$

and run the update

$$z^{k+1} = z^k - \underbrace{\nabla^2 \ell(z^k)^{-1} \nabla \ell(z^k)}_{\Delta z^k}.$$

Newton's Method via Quadratic Optimization

Observe that

$$\nabla \ell(z^k) + \nabla^2 \ell(z^k) \Delta z^k = 0$$

is the first-order necessary and sufficient condition of optimality for the quadratic program

$$\Delta z^k = \arg\min_{\Delta z} \nabla \ell(z^k)^\top \Delta z + \tfrac{1}{2} \Delta z^\top \nabla^2 \ell(z^k) \Delta z. \tag{1}$$

Thus, the k-th iteration of Newton's method can be seen as

$$z^{k+1} = z^k - \Delta z^k$$

with Δz^k solution of the quadratic program (1).

Gradient methods via quadratic optimization

Similarly to the Newton's method, a descent direction $\Delta z^k = D^k \nabla \ell(z^k)$ can be seen as the direction that minimizes at each iteration a different quadratic approximation of ℓ about z^k.

In fact, consider the quadratic approximation $\ell^k(z)$ of ℓ about z^k given by

$$\ell^k(z) = \ell(z^k) + \nabla \ell(z^k)^\top (z - z^k) + \tfrac{1}{2}(z - z^k)^\top (D^k)^{-1}(z - z^k)$$

By setting the derivative to zero, we have

$$\nabla \ell(z^k) + (D^k)^{-1}(z - z^k) = 0$$

we can calculate the minimum of $\ell^k(z)$ and set it as the next iterate z^{k+1}

$$z^{k+1} = z^k - D^k \nabla \ell(z^k)$$

namely,

$$\Delta z^k = -D^k \nabla \ell(z^k)$$

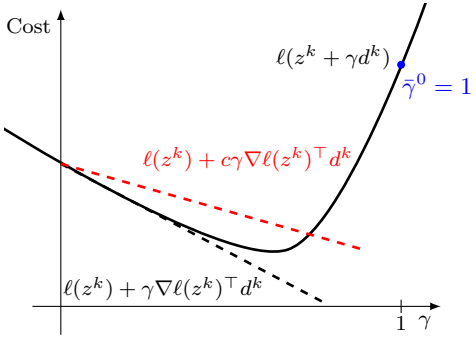
Remark Direction Δz^k can be computed by solving the quadratic approximation. This can be very helpful in optimization problems in which a gradient is not (easily) available, but it is to solve a quadratic optimization problem. It is useful also in case the domain is a compact convex set.

Gradient methods: step-size selection rules (I)

Armijo rule (Backtracking line-search)

Step-size is selected following the procedure:

1. Set $\bar{\gamma}^0 > 0$, $\beta \in (0,1)$, $c \in (0,1)$
2. While $\ell(z^k + \bar{\gamma}^i d^k) \geq \ell(z^k) + c\bar{\gamma}^i \nabla \ell(z^k)^\top d^k$:
$$\bar{\gamma}^{i+1} = \beta \bar{\gamma}^i$$
3. Set $\gamma^k = \bar{\gamma}^i$



Idea: Select γ^k such that cost is reduced while the algorithm makes progress.

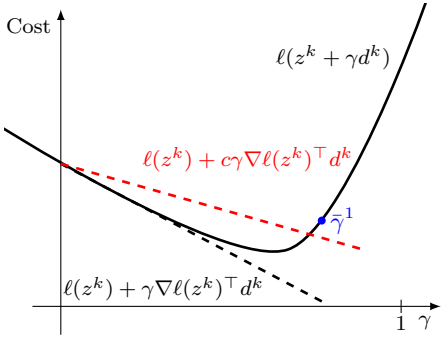
Remark Typical values are $\beta = 0.7$ and $c = 0.5$.

Gradient methods: step-size selection rules (I)

Armijo rule (Backtracking line-search)

Step-size is selected following the procedure:

1. Set $\bar{\gamma}^0 > 0$, $\beta \in (0,1)$, $c \in (0,1)$
2. While $\ell(z^k + \bar{\gamma}^i d^k) \geq \ell(z^k) + c\bar{\gamma}^i \nabla \ell(z^k)^\top d^k$:
$$\bar{\gamma}^{i+1} = \beta \bar{\gamma}^i$$
3. Set $\gamma^k = \bar{\gamma}^i$



Idea: Select γ^k such that cost is reduced while the algorithm makes progress.

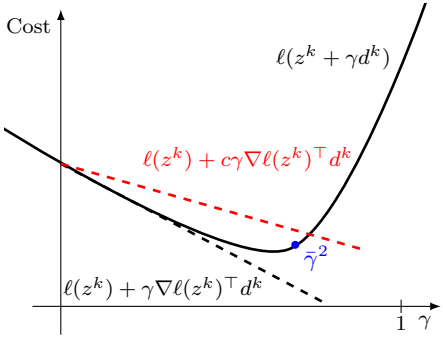
Remark Typical values are $\beta = 0.7$ and $c = 0.5$.

Gradient methods: step-size selection rules (I)

Armijo rule (Backtracking line-search)

Step-size is selected following the procedure:

1. Set $\bar{\gamma}^0 > 0$, $\beta \in (0,1)$, $c \in (0,1)$
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Idea: Select γ^k such that cost is reduced while the algorithm makes progress.

Remark Typical values are $\beta = 0.7$ and $c = 0.5$.

Gradient methods: step-size selection rules (II)

Constant step-size $\gamma^k = \gamma > 0$ for all $t = 0, 1, 2 \dots$

Diminishing step-size $\gamma^k \to 0$
It must be avoided that becomes too small to guarantee substantial progress. For this reason we require

$$\sum_{t=0}^{\infty} \gamma^k = \infty.$$

Remark The associated convergence rate tends to be slow.

Typical choice: γ^k chosen such that

$$\sum_{t=0}^{\infty} \gamma^k = \infty \quad \text{and} \quad \sum_{t=0}^{\infty} \left(\gamma^k\right)^2 < \infty,$$

e.g., $\gamma^k = \frac{1}{k^\alpha}$ with $\frac{1}{2} < \alpha \leq 1$.

Gradient methods: convergence result (Armijo)

Proposition (convergence with Armijo step-size)
Let $\{z^k\}$ be a sequence generated by a gradient method $z^{k+1} = z^k - \gamma^k D^k \nabla \ell(z^k)$ with $d_1 I \leq D^k \leq d_2 I$, $d_1 > 0$, $d_2 > 0$. Assume that γ^k is chosen by the Armijo rule and $\ell(z)$ be continuously differentiable. Then, every limit point $\bar{z}$ of the sequence $\{z^k\}$ is a stationary point, i.e., $\nabla \ell(\bar{z}) = 0$.

Remark Recall that a vector $z \in \mathbb{R}^d$ is a limit point of a sequence $\{z^k\}$ in $\mathbb{R}^d$ if there exists a subsequence of $\{z^k\}$ that converges to z.

Gradient methods: convergence result (constant/diminishing)

Proposition (convergence with constant or diminishing step-size)
Let $\{z^k\}$ be a sequence generated by a gradient method $z^{k+1} = z^k - \gamma^k D^k \nabla \ell(z^k)$ with $d_1 I \leq D^k \leq d_2 I$, $d_1 > 0$, $d_2 > 0$. Assume that for some $L > 0$

$$\|\nabla \ell(z) - \nabla \ell(y)\| \leq L\|z - y\|, \qquad \forall z, y \in \mathbb{R}^d.$$

Assume either

1. $\gamma^k = \gamma > 0$ sufficiently small, or
2. $\gamma^k \to 0$ and $\sum_{t=0}^{\infty} \gamma^k = \infty$.

Then, every limit point $\bar{z}$ of the sequence $\{z^k\}$ is a stationary point, i.e., $\nabla \ell(\bar{z}) = 0$.

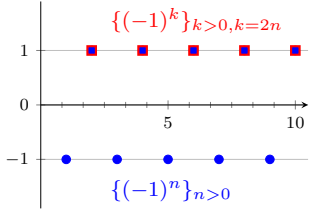
Remark Check if the following problem $\min_{z \in \mathbb{R}^d} c^\top z$, with $c \in \mathbb{R}^\top$ satisfies the assumptions. □

Gradient methods: remarks

Remark The propositions do not guarantee that the sequence converges (to a point) and not even existence of limit points. Then either $\ell(z^k) \to -\infty$ or $\ell(z^k)$ converges to a finite value and $\nabla \ell(z^k) \to 0$. In the second case, one can show that any subsequence $\{z^{k_p}\}$ converges to some stationary point $\bar{z}$ satisfying $\nabla \ell(\bar{z}) = 0$. □

Remark Existence of minima can be guaranteed by excluding $\ell(z^k) \to -\infty$ via suitable assumptions. Assume, e.g., ℓ coercive (or radially unbounded), i.e., $\lim_{\|z\| \to \infty} \ell(z) = \infty$. □

Remark For general (nonconvex) problems, if coercivity is assumed, only convergence (of subsequences) to stationary points (i.e., to $\bar{z}$ with $\nabla \ell(\bar{z}) = 0$) can be proven. □



Gradient methods: remarks

Remark The propositions do not guarantee that the sequence converges (to a point) and not even existence of limit points. Then either $\ell(z^k) \to -\infty$ or $\ell(z^k)$ converges to a finite value and $\nabla \ell(z^k) \to 0$. In the second case, one can show that any subsequence $\{z^{k_p}\}$ converges to some stationary point $\bar{z}$ satisfying $\nabla \ell(\bar{z}) = 0$. □

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Remark For general (nonconvex) problems, if coercivity is assumed, only convergence (of subsequences) to stationary points (i.e., to $\bar{z}$ with $\nabla \ell(\bar{z}) = 0$) can be proven. □

Remark For convex programs, assuming coercivity, convergence (of subsequences) to global minima is guaranteed since necessary conditions of optimality are also sufficient. □

Example: Gradient Method applied to Quadratic Programs

Quadratic program

$$\min_{z \in \mathbb{R}^d} \ \frac{1}{2} z^T Q z + q^T z$$

with $Q = Q^T \geq 0 \in \mathbb{R}^{d \times d}$ and $q \in \mathbb{R}^d$.

Steepest descent method

$$z^{k+1} = z^k - \gamma^k \nabla \ell(z^k)$$

for a Quadratic Program is given by

$$z^{k+1} = z^k - \gamma^k (Q z^k + q).$$

Optimization (constrained) over convex sets: optimality conditions and algorithms

Optimization over convex sets

Consider the optimization problem

$$\min_{z \in Z} \ell(z),$$

where $Z \subset \mathbb{R}^d$ is nonempty, convex and closed, and ℓ is continuously differentiable on Z.
Remark Recall that a set Z is closed if it contains all of its limit points.

IMPORTANT
The function ℓ can be nonconvex.

Optimality Conditions

If a point $z^* \in Z$ is a local minimum of $\ell(z)$ over Z, then

$$\nabla \ell(z^*)^\top (\bar{z} - z^*) \geq 0 \qquad \forall \bar{z} \in Z$$

Optimization over convex sets

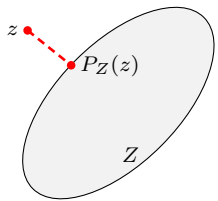
Projection over a convex set

Given a point $z \in \mathbb{R}^d$ and a closed convex set Z, it can be shown that

$$P_Z(z) := \arg\min_{y \in Z} \|y - z\|^2$$

exists and is unique.

The point $P_Z(z)$ is called the projection of z on Z.



Question: can you prove existence and uniqueness of projection point?

Projected gradient

Gradient methods can be generalized to optimization over convex sets.

Projected gradient

$$z^{k+1} = P_Z\Big(z^k - \gamma^k \nabla \ell(z^k)\Big).$$

The algorithm is based on the idea of generating at each t feasible points (i.e., belonging to Z) that give a descent in the cost.

The analysis follows similar arguments to the one of unconstrained gradient methods.

Feasible direction method

Find $\tilde{z} \in \mathbb{R}^d$ such that

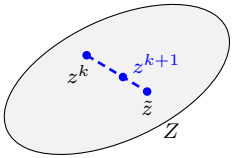
$$\tilde{z} = \underset{z \in Z}{\operatorname{argmin}} \ \cancel{\ell(z^k)} + \nabla \ell(z^k)^\top (z - z^k) + \tfrac{1}{2}(z - z^k)^\top (z - z^k)$$

Update the solution

$$z^{k+1} = z^k + \gamma^k \underbrace{(\tilde{z} - z^k)}_{\text{feasible direction}}$$

Note: For γ^k sufficiently small, $z^{k+1} \in Z$.

$$\begin{aligned} z^{k+1} &= z^k + \gamma^k (\tilde{z} - z^k) \\ &= (1 - \gamma^k) z^k + \gamma^k \tilde{z} \end{aligned}$$



Example: Projected Gradient Method for Quadratic Programs

Quadratic program

$$\min_{z \in Z} \frac{1}{2} z^T Q z + q^T z$$

with $Q = Q^T \geq 0 \in \mathbb{R}^{d \times d}$ and $q \in \mathbb{R}^d$.

Projected gradient method

$$z^{k+1} = P_Z\Big(z^k - \gamma^k \nabla \ell(z^k)\Big)$$

is given by

$$z^{k+1} = P_Z\Big(z^k - \gamma^k (Q z^k + q)\Big).$$

Examples of cost functions

$$\ell(z) = z^2$$

$$\ell(z) = z^4 + 2z^3 - 12z^2 - 2z + 6$$

$$\ell(z) = e^{z_1+3z_2-0.1} + e^{z_1-3z_2-0.1} + e^{-z_1-0.1}$$

$$\ell(z) = \tfrac{1}{2} z^\top Q z + r^\top z$$

Barrier function strategy for inequality constraints

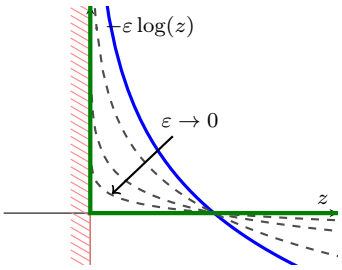
Consider the inequality constrained optimization problem

$$\min_{z \in \mathbb{R}^d} \ell(z)$$
$$\text{subj.to } g_j(z) \leq 0 \qquad j \in \{1, \dots, r\}$$

Inequality constraints can be relaxed and embedded in the cost function by means of a barrier function $-\varepsilon \log(z)$.

The resulting unconstrained problem reads as

$$\min_{z \in \mathbb{R}^d} \ell(z) + \varepsilon \sum_{j=1}^{r} -\log(-g_j(z))$$



Implementation: every few iterations shrink the barrier parameters ε.

Constrained optimization (equality and inequality constraints): optimality conditions

Constrained optimization problem set-up

$$
\begin{aligned}
&\min_{z \in \mathbb{R}^d} \ell(z) \\
\text{subj.to } & g_j(z) \leq 0 \qquad j \in \{1, \dots, r\} \\
& h_i(z) = 0 \qquad i \in \{1, \dots, m\}
\end{aligned}
$$

Active constraints and regular points

Set of active inequality constraints

For a point z, the set of active inequality constraints at z is $A(z) = \{j \in \{1, \dots, r\} \mid g_j(z) = 0\}$.

Regular point

A point z is regular if the vectors $\nabla h_i(z)$, $i \in \{1, \dots, m\}$, and $\nabla g_j(z)$, $j \in A(z)$, are linearly independent.

Lagrangian function

In order to state the first-order necessary conditions of optimality for (equality and inequality) constrained problems it is useful to introduce the Lagrangian function

$$L(z, \mu, \lambda) = \ell(z) + \sum_{j=1}^{r} \mu_j g_j(z) + \sum_{i=1}^{m} \lambda_i h_i(z)$$

with $z \in \mathbb{R}^d$, $\mu \in \mathbb{R}^r$ and $\lambda \in \mathbb{R}^m$.

Remark μ_j and λ_i can be seen as prices for violating the associated constraint. □

Remark We are relaxing the constrained problem, but we cannot minimize the Lagrangian. □

Karush-Kuhn-Tucker (KKT) necessary conditions

Let $z^\star$ be a regular local minimum of

$$\begin{aligned} &\min_{z \in \mathbb{R}^d} \ell(z) \\ \text{subj.to } &g_j(z) \leq 0 \qquad j \in \{1, \dots, r\} \\ &h_i(z) = 0 \qquad i \in \{1, \dots, m\} \end{aligned}$$

where ℓ, g_j and h_i are C^1.

Karush-Kuhn-Tucker (KKT) necessary conditions

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where ℓ, g_j and h_i are C^1.

Then $\exists$ unique $\mu_j^\star$ and $\lambda_i^\star$, called Lagrange multipliers, s.t.

$$\begin{aligned} \nabla_1 L(z^\star, \mu^\star, \lambda^\star) &= 0 \\ \mu_j^\star &\geq 0 \\ \mu_j^\star \, g_j(z^\star) &= 0 \qquad j \in \{1, \dots, r\} \end{aligned}$$

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Remark If ℓ, h_i and g_j twice differentiable, second order conditions.

Karush-Kuhn-Tucker (KKT) necessary conditions

Let $z^\star$ be a regular local minimum of

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Notation Points satisfying the KKT necessary conditions of optimality are referred to as KKT points. They are the counterpart of stationary points in constrained optimization.

Remarks on Karush-Kuhn-Tucker (KKT) necessary conditions

Remark The condition

$$\nabla_1 L(z^\star, \mu^\star, \lambda^\star) = 0$$

can be explicitly written as

$$\nabla \ell(z^\star) + \sum_{j=1}^{r} \mu_j^\star \nabla g_j(z^\star) + \sum_{i=1}^{m} \lambda_i^\star \nabla h_i(z^\star) = 0$$

Remark The condition $\mu_j^\star \, g_j^\star(z^\star) = 0$, $j \in \{1, \ldots, r\}$, is called complementary slackness.

Karush-Kuhn-Tucker (KKT) necessary conditions (second order)

Let $z^\star$ be a regular local minimum of

$$\begin{aligned} &\min_{z \in \mathbb{R}^d} \ell(z) \\ &\text{subj.to } g_j(z) \leq 0 \qquad j \in \{1, \dots, r\} \\ &\qquad\quad h_i(z) = 0 \qquad i \in \{1, \dots, m\} \end{aligned}$$

where ℓ, g_j and h_i are C^1.

Then $\exists$ unique $\mu_j^\star$ and $\lambda_i^\star$, called Lagrange multipliers, s.t.

$$\nabla_1 \mathcal{L}(z^\star, \mu^\star, \lambda^\star) = 0, \qquad \mu_j^\star \geq 0, \qquad \mu_j^\star \, g_j(z^\star) = 0 \qquad j \in \{1, \dots, r\}$$

Karush-Kuhn-Tucker (KKT) necessary conditions (second order)

Let $z^\star$ be a regular local minimum of

$$\begin{aligned} &\min_{z\in\mathbb{R}^d} \ell(z) \\ \text{subj.to } &g_j(z) \leq 0 \qquad j \in \{1,\dots,r\} \\ &h_i(z) = 0 \qquad i \in \{1,\dots,m\} \end{aligned}$$

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Then $\exists$ unique $\mu_j^\star$ and $\lambda_i^\star$, called Lagrange multipliers, s.t.

$$\nabla_1 \mathcal{L}(z^\star, \mu^\star, \lambda^\star) = 0, \qquad \mu_j^\star \geq 0, \qquad \mu_j^\star \, g_j(z^\star) = 0 \qquad j \in \{1,\dots,r\}$$

Moreover, if ℓ, g_j and h_i are C^2 it holds

$$y^\top \nabla_{11}^2 \mathcal{L}(z^\star, \mu^\star, \lambda^\star) y \geq 0$$

for all $y \in \mathbb{R}^d$ such that

$$\nabla h_i(z)^\top y = 0, \quad i \in \{1,\dots,m\}, \qquad \nabla g_j(z)^\top y = 0, \quad j \in A(z) \quad \text{(i.e. } j \in \{1,\dots,r\} \text{ s.t. } g_j(z) = 0)$$

Quadratic Programming (constrained)

Let us consider quadratic optimization problems with linear equality constraints.

Quadratic program

$$\begin{aligned} \min_{z \in \mathbb{R}^d} \quad & \frac{1}{2} z^T Q z + q^T z \\ \text{subj.to} \quad & Az = b \end{aligned}$$

with $Q = Q^T \in \mathbb{R}^{d \times d}$, $q \in \mathbb{R}^d$, $A \in \mathbb{R}^{m \times d}$ and $b \in \mathbb{R}^m$.

Quadratic Programming (constrained): optimality conditions

Let us introduce the Lagrangian of the constrained optimization problem, i.e.,

$$\mathcal{L}(z, \lambda) = \frac{1}{2} z^T Q z + q^T z + \lambda^\top (Az - b) \qquad \lambda \in \mathbb{R}^m$$

First-order necessary condition for optimality: if $z^\star$ is a minimum then there exists $\lambda^\star$ such that

$$\begin{bmatrix} \nabla_1 \mathcal{L}(z^\star, \lambda^\star) \\ \nabla_2 \mathcal{L}(z^\star, \lambda^\star) \end{bmatrix} = 0 \implies \begin{bmatrix} Qz^\star + q + A^\top \lambda^\star \\ Az^\star - b \end{bmatrix} = 0$$

Quadratic Programming (constrained): optimality conditions

Second-order necessary condition for optimality: if $z^\star$ is a minimum then

$$y^\top \nabla^2_{11} \mathcal{L}(z^\star, \lambda^\star) y = y^\top Q y \geq 0$$

for all $y \in \mathbb{R}^d$ such that

$$\nabla h_i(z)^\top y = 0 \qquad i \in \{1, \dots, m\}$$
$$\implies A^\top y = 0$$

namely, for all $y \in \mathbb{R}^d$ in the null-space of $A^\top$.

Constrained optimization (equality and inequality constraints): optimization algorithms

Optimality conditions for equality constrained problems (refresh)

Consider the equality-constrained minimization problem:

$$\begin{aligned} \min_{z \in \mathbb{R}^d} \quad & \ell(z) \\ \text{subj.to} \quad & h(z) = 0 \end{aligned}$$

where $\ell : \mathbb{R}^d \to \mathbb{R}$ and $h : \mathbb{R}^d \to \mathbb{R}^m$.

The first-order necessary conditions of optimality require the solution of

$$\begin{aligned} \nabla \ell(z) + \nabla h(z) \lambda &= 0 \\ h(z) &= 0 \end{aligned}$$

which can be written as

$$\nabla L(z, \lambda) = \begin{bmatrix} \nabla \ell(z) + \nabla h(z) \lambda \\ h(z) \end{bmatrix} = 0.$$

Newton's method for equality constrained problems

KKT points can be found by solving a root finding problem in variables z, λ with $r(z, \lambda) = \nabla L(z, \lambda)$.

Newton's method for this root finding problem reads as

$$\begin{bmatrix} z^{k+1} \\ \lambda^{k+1} \end{bmatrix} = \begin{bmatrix} z^k \\ \lambda^k \end{bmatrix} + \begin{bmatrix} \Delta z^k \\ \Delta \lambda^k \end{bmatrix}$$

with Δz^k, $\Delta \lambda^k$ solution of

$$\nabla^2 L(z^k, \lambda^k) \begin{bmatrix} \Delta z \\ \Delta \lambda \end{bmatrix} = -\nabla L(z^k, \lambda^k)$$

where

$$\nabla^2 L(z^k, \lambda^k) := \begin{bmatrix} B^k & \nabla h(z^k) \\ \nabla h(z^k)^\top & 0 \end{bmatrix} \qquad \nabla L(z^k, \lambda^k) := \begin{bmatrix} \nabla \ell(z^k) + \nabla h(z^k) \lambda^k \\ h(z^k) \end{bmatrix}$$

and $B^k := \nabla^2_{11} L(z^k, \lambda^k)$.

Newton's method for equality constrained problems

Namely, search for Δz^k, $\Delta \lambda^k$ solution of

$$\begin{bmatrix} B^k & \nabla h(z^k) \\ \nabla h(z^k)^\top & 0 \end{bmatrix} \begin{bmatrix} \Delta z \\ \Delta \lambda \end{bmatrix} = - \begin{bmatrix} \nabla \ell(z^k) + \nabla h(z^k) \lambda^k \\ h(z^k). \end{bmatrix}$$

Thus, Δz^k, $\Delta \lambda^k$ can be obtained as solution of a linear system of equations in the variables Δz, $\Delta \lambda$.

Newton's method via Quadratic Programming

The linear system of equations can be rewritten as

$$B^k \Delta z^k + \nabla h(z^k) \Delta \lambda^k = -\nabla \ell(z^k) - \nabla h(z^k) \lambda^k$$
$$\nabla h(z^k)^\top \Delta z^k = -h(z^k).$$

and equivalently as

$$\nabla \ell(z^k) + B^k \Delta z^k + \nabla h(z^k) \lambda^{k+1} = 0$$
$$h(z^k) + \nabla h(z^k)^\top \Delta z^k = 0.$$

We can observe that

$$\nabla \ell(z^k) + B^k \Delta z + \nabla h(z^k) \lambda = 0$$
$$h(z^k) + \nabla h(z^k)^\top \Delta z = 0$$

are the necessary and sufficient optimality conditions for the Quadratic Program (QP)

$$\min_{\Delta z} \ \nabla \ell(z^k)^\top \Delta z + \tfrac{1}{2} \Delta z^\top B^k \Delta z$$
$$\text{subj.to} \ \ h(z^k) + \nabla h(z^k)^\top \Delta z = 0.$$

Therefore, in the Newton's update, we can obtain $(\Delta z^k, \lambda^{k+1})$ by solving this QP.

Newton's Method via Sequential Quadratic Programming (SQP)

Start from a tentative solution z^0

For $k = 0, 1, \dots$ (up to convergence)

1. Compute $\nabla \ell(z^k), B^k, \nabla h(z^k)$
2. Obtain $(\Delta z^k, \Delta \lambda^k_{\text{QP}})$ from

$$\begin{aligned} \Delta z^k = \underset{\Delta z}{\text{argmin}} \;\; & \nabla \ell(z^k)^\top \Delta z + \tfrac{1}{2} \Delta z^\top B^k \Delta z \\ \text{subj.to} \;\; & h(z^k) + \nabla h(z^k)^\top \Delta z = 0. \end{aligned} \tag{2}$$

   with $\Delta \lambda^k_{\text{QP}}$ the Lagrange multiplier associated to the optimal solution of (2).

3. Update

$$\begin{aligned} z^{k+1} &= z^k + \Delta z^k \\ \lambda^{k+1} &= \Delta \lambda^k_{\text{QP}} \end{aligned}$$

Stepsize selection for constrained optimization (I): Merit function

Goal: Choose γ^k such that direction Δz^k simultaneously

i) decreases cost $\ell(z)$

ii) reduces constraint violation, i.e., $\|h(z)\|$

Idea: Measure progress, i.e., select stepsize γ^k, evaluating a *merit function*.

L_1 merit function: Consider the optimization problem with cost $\ell(z)$ and equality constraint $h(z) = 0$, the associated L_1 *merit function* reads

$$M_1(z; s) = \ell(z) + s\|g(z)\|_1$$

where, $s \in \mathbb{R}$ parameter and $\|\cdot\|_1$ is the 1-norm, i.e., for $z \in \mathbb{R}^d$, $\|z\|_1 = \sum_{i=1}^{d} |z_i|$.

Proposition (exact merit function) Call λ^* the optimal Lagrange multiplier associated to the original constrained optimization problem. Then, if $s > \max_i |\lambda_i^*|$, $i = 1, \ldots, m$, the merit function $M_1(z; s)$ is *exact*, i.e., any solution (z^*, λ^*) of the original optimization problem is a local minimizer of $M_1(z; s)$.

Stepsize selection for constrained optimization (II): Armijo rule

Armijo rule (Backtracking line-search) with M_1 merit function

Step-size is selected following the procedure:

1. Set $\bar{\gamma}^0 > 0$, $\beta \in (0,1)$, $c \in (0,1)$
2. While $M_1(z^k + \bar{\gamma}^i d^k; s) \geq M_1(z^k; s) + c\bar{\gamma}^i \overbrace{\left(\nabla \ell(z^k)^\top d^k - s\|h(z^k)\|_1\right)}^{\text{derivative of } M_1(z^k;s) \text{ along } d^k}$:
$$\bar{\gamma}^{i+1} = \beta \bar{\gamma}^i$$
3. Set $\gamma^k = \bar{\gamma}^i$

Good practice: to move towards exact merit function, adapt s at each iteration, i.e., set $s^k > \|\lambda^k\|_\infty$

Sequential Quadratic Programming (SQP)

Start from a tentative solution z^0

For $k = 0, 1, \dots$ (up to convergence)

1. Compute $\nabla \ell(z^k), B^k, \nabla h(z^k)$
2. Obtain $(\Delta z^k, \Delta \lambda^k_{\text{QP}})$ from

$$\begin{aligned} \Delta z^k = \underset{\Delta z}{\text{argmin}} \;\; & \nabla \ell(z^k)^\top \Delta z + \tfrac{1}{2} \Delta z^\top B^k \Delta z \\ \text{subj.to} \;\; & h(z^k) + \nabla h(z^k)^\top \Delta z = 0. \end{aligned} \tag{3}$$

   with $\Delta \lambda^k_{\text{QP}}$ the Lagrange multiplier associated to the optimal solution of (3).

3. Choose γ^k using Armijo's rule on *merit function* $M_1(z^k + \gamma \Delta z^k; s)$
4. Update

$$\begin{aligned} z^{k+1} &= z^k + \gamma^k \Delta z^k \\ \lambda^{k+1} &= (1 - \gamma^k) \lambda^k + \gamma^k \Delta \lambda^k_{\text{QP}} \end{aligned}$$

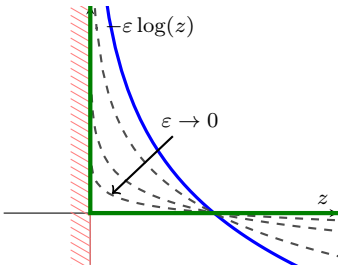
Barrier function strategy for inequality constraints

Consider the inequality constrained optimization problem

$$\begin{aligned} &\min_{z\in\mathbb{R}^d} \ell(z) \\ &\text{subj.to } g_j(z) \leq 0 \qquad j \in \{1,\dots,r\} \\ &\qquad h(z) = 0 \end{aligned}$$

Inequality constraints can be embedded in the cost function by means of a barrier function $-\varepsilon \log(z)$
The resulting unconstrained problem reads as

$$\begin{aligned} &\min_{z\in\mathbb{R}^d} \ell(z) + \varepsilon \sum_{j=1}^{r} -\log(-g_j(z)) \\ &\qquad h(z) = 0 \end{aligned}$$



Implementation: every few iterations shrink the barrier parameters ε.

Concluding Remarks on Notation

Warning on notation in Optimization and Optimal Control

Notation z

We denote $z \in \mathbb{R}^d$ the generic optimization variable of a generic optimization problem, e.g.

$$\min_{z \in \mathbb{R}^d} \ell(z),$$

where $\ell : \mathbb{R}^d \to \mathbb{R}$ cost function.

Notation z^k

We denote $z^k \in \mathbb{R}^d$ the solution estimate at iteration $k \in \mathbb{N}$ of an optimization algorithm for the optimization problem.

Warning on notation in Optimization and Optimal Control

Notation x_t

Consider a discrete-time system in standard control notation $x(t+1) = f_t(x(t), u(t))$ with $t \in \mathbb{N}$.
We denote $x_t := x(t)$ and $u_t := u(t)$ the state and the input respectively, so that the dynamics is written as

$$x_{t+1} = f_t(x_t, u_t)$$

Notation $(\mathbf{x}, \mathbf{u})$, $(\mathbf{x}^k, \mathbf{u}^k)$, (x_t^k, u_t^k)

The decision vector in optimal control problems is $(\mathbf{x}, \mathbf{u})$ and we will denote $(\mathbf{x}^k, \mathbf{u}^k)$ the solution estimate at iteration k of an optimal control algorithm.

We will also use (x_t^k, u_t^k) to denote the vector component of $(\mathbf{x}^k, \mathbf{u}^k)$ associated to the state and input at time t.